

# Sputtered AlN buffer layer for low loss crystalline AlN on sapphire integrated photonics

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**Abstract** To fully exploit the broad transparency window of crystalline AlN through nonlinear optics, both dispersion engineering and low optical losses are required. We present thick regrown AlN waveguides enabling three-octave supercontinuum generation and show how a sputtered AlN buffer suppresses void-induced scattering, establishing a pathway toward low-loss visible-to-mid-infrared integrated photonics. ©2026 The Author(s)

## Introduction

Aluminum nitride (AlN) has emerged as a promising nitride for integrated photonics owing to its unique combination of a wide transparency window extending from the deep ultraviolet (~200 nm) to the mid-infrared (mid-IR), a strong second-order nonlinear susceptibility, and an intrinsic Kerr nonlinearity. These properties make crystalline AlN-on-sapphire attractive for applications including frequency conversion, integrated frequency comb generation, visible photonics, quantum technologies, and precision sensing [1-5].

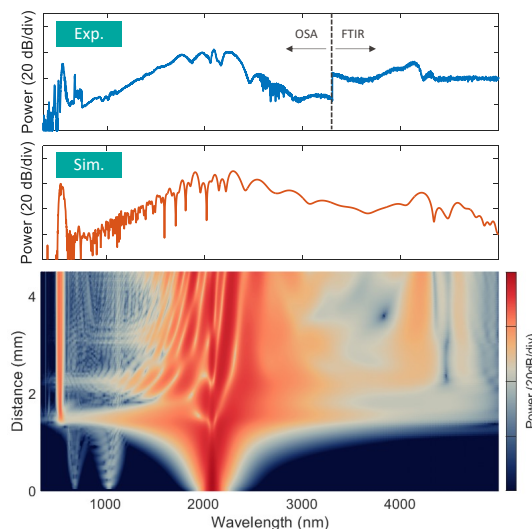
To exploit the exceptional transparency range of AlN through nonlinear optical processes, both precise dispersion engineering and low optical losses are required. In recent work, we developed an epitaxial regrowth process enabling crystalline AlN waveguides with thicknesses beyond those available in standard commercial platforms [2]. The resulting 1.3  $\mu\text{m}$ -thick waveguides provide increased flexibility for dispersion engineering and support broadband supercontinuum generation spanning more than three octaves, from 550 nm to 4.5  $\mu\text{m}$  [2]. While the overall spectral evolution agrees well with numerical simulations, the experimentally observed visible power remains lower than predicted, suggesting the presence of wavelength-dependent loss mechanisms.

In this work, we investigate the origin of these losses and their impact on crystalline AlN photonic devices. By comparing fully MOVPE-grown, commercial, and hybrid approach where a thin AlN layer is first sputtered before the MOVPE growth, we identify void-induced scattering as a key limiting mechanism for short-wavelength operation. We show that introducing a thin sputtered AlN buffer layer prior to epitaxial growth suppresses void formation and establishes a pathway toward low-loss photonic circuits spanning the visible to the mid-IR [1].

Using this hybrid crystalline AlN-on-sapphire

platform, we demonstrate intrinsic quality factors up to  $2 \times 10^6$  and propagation losses below 0.2 dB/cm at 1550 nm [1]. While these metrics are already competitive with the best commercial crystalline AlN templates, the complete suppression of nanoscale voids is expected to provide substantially larger benefits at visible and ultraviolet wavelengths, where volumetric scattering becomes increasingly important.

## Thick Crystalline AlN for Broadband Nonlinear Optics



**Fig. 1:** Supercontinuum generation 2.5  $\mu\text{m} \times 1.3 \mu\text{m}$  waveguide after regrowth on commercial AlN layer, pumped at 2.1  $\mu\text{m}$ ; Bottom: simulated evolution of the spectrum upon propagation distance; Middle: simulated output spectrum; Top: Experimentally measured output spectrum.

One of the major challenges in crystalline AlN photonics is the limited thickness of commercially available epilayers, which constrains dispersion engineering and therefore the achievable nonlinear optical performance. To address this limitation, we developed an epitaxial regrowth approach allowing the realization of crystalline AlN layers thicker than the standard platform

while preserving excellent optical quality [2].

The increased thickness substantially expands the accessible dispersion-engineering space and enables operation in regimes inaccessible with conventional waveguide geometries. Using 1.3  $\mu\text{m}$ -thick crystalline AlN waveguides, efficient dispersive-wave generation was achieved simultaneously in the visible and mid-infrared spectral regions. Pumping near 2  $\mu\text{m}$  resulted in broadband supercontinuum generation spanning more than three octaves, extending from 550 nm to 4.5  $\mu\text{m}$  [2] (Fig.1). The generated spectra demonstrate the unique capability of crystalline AlN to combine broad transparency with simultaneous  $\chi^{(2)}$  and  $\chi^{(3)}$  nonlinear interactions on a single integrated platform.

The experimental spectra show excellent agreement with numerical simulations regarding dispersive-wave positions and overall spectral evolution. However, the experimentally measured power in the visible spectral range remains lower than predicted. This observation suggests that wavelength-dependent loss mechanisms, only weakly revealed by conventional telecom-band characterization, may play a critical role in determining the ultimate performance of broadband nonlinear devices.

### Understanding and Mitigating Optical Losses in Crystalline AlN

The discrepancy observed between simulated and measured visible supercontinuum power motivated a systematic investigation of material-related losses in crystalline AlN-on-sapphire photonic structures

To identify the origin of these losses, three different classes of crystalline AlN epilayers were investigated: a fully MOVPE-grown layer (Sample A), a commercial AlN template (Sample B), and a hybrid structure combining a sputtered AlN buffer layer with MOVPE overgrowth (Sample C). Figure 2 summarizes the corresponding material

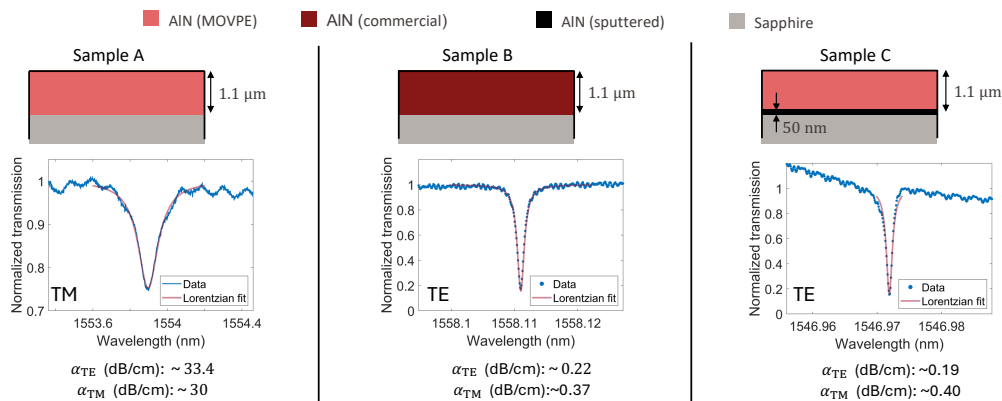
structures together with representative microresonator measurements and extracted propagation losses.

Cross-sectional scanning electron microscopy reveals a high density of vertically elongated nanoscale voids at the AlN-sapphire interface in the fully MOVPE-grown material. Commercial AlN templates exhibit significantly fewer and smaller voids, while no voids are observed in the hybrid structure employing the sputtered AlN buffer layer.

The origin of these differences can be traced to the early stages of epitaxial growth. Conventional MOVPE growth on sapphire typically relies on a three-dimensional growth regime that promotes strain relaxation and crack suppression but can also lead to the formation of interfacial voids during island coalescence. In contrast, the sputtered AlN buffer layer provides a continuous polycrystalline template prior to MOVPE overgrowth, suppressing the formation of these voids and enabling the realization of void-free crystalline AlN epilayers [1].

The optical performance closely follows the observed material quality. Waveguides fabricated on fully MOVPE-grown AlN exhibit propagation losses exceeding 30 dB/cm at telecom wavelengths, confirming the detrimental impact of a high density of voids. In contrast, both commercial and hybrid AlN support low-loss photonic devices, with measured TE propagation losses of 0.22 dB/cm and 0.19 dB/cm, respectively. Corresponding resonators exhibit intrinsic quality factors of approximately  $1.7 \times 10^6$  for the commercial material and up to  $2 \times 10^6$  for the hybrid platform. The relatively small difference observed at 1550 nm indicates that the residual void density present in commercial AlN only weakly impacts telecom-band operation, in particular in these relatively thick layers.

At first glance, these results suggest that residual voids have little practical impact on



**Fig. 2:** Right: sample A material stack of full MOVPE AlN; Middle: sample B material stack of commercial AlN; Left: sample C material stack of MOVPE AlN on a thin sputtered AlN layer and their corresponding transmission spectrum of TM or TE resonance from fabricated microrings.

photonic performance. However, finite-difference time-domain simulations reveal a markedly different picture at shorter wavelengths. Because volumetric scattering scales approximately as  $\lambda^{-4}$ , the influence of residual voids increases rapidly toward the visible and ultraviolet regions. This effect is further amplified in thinner waveguides, where a larger fraction of the optical mode overlaps with the defect-rich region close to the AlN-sapphire interface.

Consequently, although commercial and hybrid AlN exhibit comparable performance at telecom wavelengths, the void-free hybrid platform is expected to provide substantially lower losses and significantly improved device performance in visible and ultraviolet photonic circuits. These results provide a plausible explanation for the reduced visible power observed in broadband nonlinear experiments and highlight the importance of considering material-related scattering mechanisms when designing photonic devices intended to operate across the entire transparency window of AlN.

## Conclusion

Fully exploiting the exceptional transparency and nonlinear properties of crystalline AlN requires both dispersion engineering and careful control of wavelength-dependent optical losses. Epitaxial regrowth enables thick crystalline AlN waveguides supporting broadband supercontinuum generation over more than three octaves. A systematic investigation of material quality reveals the critical role of void-induced scattering in AlN-on-sapphire photonics. While commercial crystalline AlN already exhibits excellent telecom-band performance, simulations indicate that residual voids increasingly limit operation toward shorter wavelengths. The introduction of a sputtered AlN buffer layer suppresses void formation entirely while preserving low-loss telecom performance, establishing a pathway toward photonic integrated circuits spanning the visible, near-infrared, and mid-infrared spectral regions.

- [1] S. Brunetta, S. Sbarra, B. S. Y. Loke, J.-F. Carlin, N. Grandjean, C.-S. Brès, and R. Butté, "Sputtered AlN Buffer Layer for Low-Loss Crystalline AlN-on-Sapphire Integrated Photonics", *ACS Photonics*, vol. 13, pp. 1388–1398, 2026, DOI: 10.1021/acsp Photonics.5c02661.
- [2] S. Sbarra, S. Brunetta, P. A. Demongodin, J.-F. Carlin, N. Grandjean, R. Butté, and C.-S. Brès, "Three-octave supercontinuum generation in thick crystalline aluminum nitride waveguides", *Optics Letters*, vol. 50, no. 22, pp. 7147–7150, 2025, DOI: 10.1364/OL.577343.
- [3] X. Liu, A. W. Bruch, Z. Gong, J. Lu, and H. X. Tang, "Ultra-high-Q UV microring resonators based on a single-crystalline AlN platform", *Optica*, vol. 5, no. 10, pp. 1279–1282, 2018, DOI: 10.1364/OPTICA.5.001279.
- [4] H. Jung, C. Xiong, K. Y. Fong, X. Zhang, and H. X. Tang, "Optical frequency comb generation from aluminum nitride microring resonator", *Optics Letters*, vol. 38, no. 15, pp. 2810–2813, 2013, DOI: 10.1364/OL.38.002810.
- [5] X. Liu, A. W. Bruch, J. Lu, J. Surya, L. Zhang, A. V. Klenner, Y. Liu, N. Sinclair, S. A. Bhave, and H. X. Tang, "Integrated high quality factor aluminum nitride microresonators", *Optica*, vol. 5, no. 10, pp. 1279–1282, 2018, DOI: 10.1364/OPTICA.5.001279.